

Lecture 12

Hedonics and Real Estate Markets

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AEM 4510

Roadmap

- What can we use to infer the demand for environmental goods?
- What do housing prices tell us?
- What is the demand for hazardous waste? (Greenstone and Gallagher, 2008)

Hedonic valuation

Revealed preference approaches

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Is there a way we can reveal the value of these goods?

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This change in price can tell us something about how people value the change in the environmental good

Revealed preference approaches: example

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What does this price change mean?

Hedonics: Property value models

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- Rooms
- Bathrooms
- School quality
- **Environmental quality**

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Homes located in pristine areas are likely to be more valuable than identical homes located near toxic facilities

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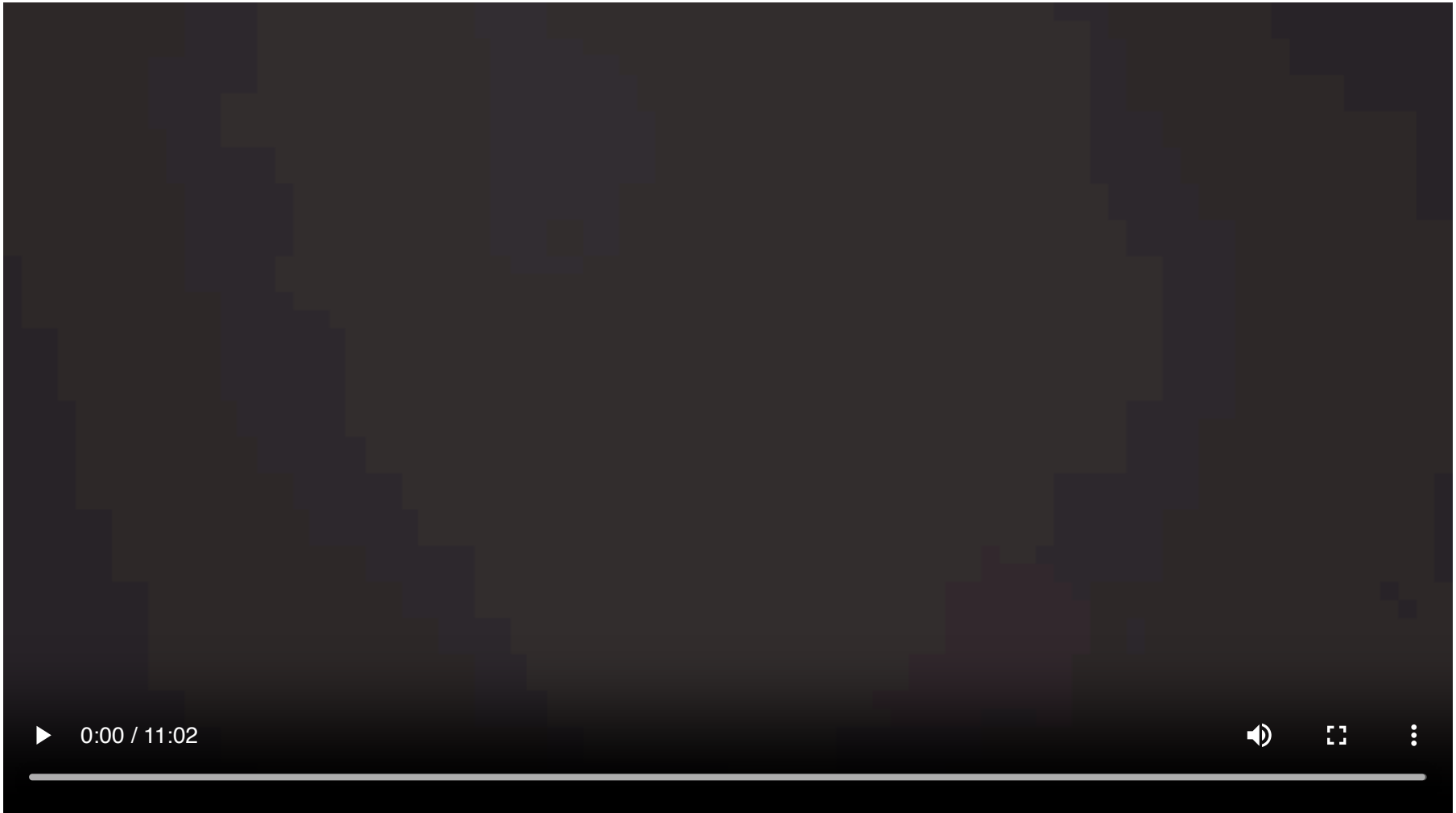
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e.g. homes in better school districts are typically more expensive

BCA of Superfund

Superfund



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By 2005: \$35 billion in federal funding has been spent at roughly 800 sites

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How do we do it?

Superfund

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Lets get some intuition for how housing markets reveal the value of environmental goods

Hedonics intuition: L.O. Taylor, 2003

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At the current equilibrium price of \$200,000 per house, all 200 hundred homes on either lake are equally preferred

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Lake A prices **increase** to bring the market back into equilibrium

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Real world considerations: houses aren't going to be perfectly comparable, etc

The hedonic model

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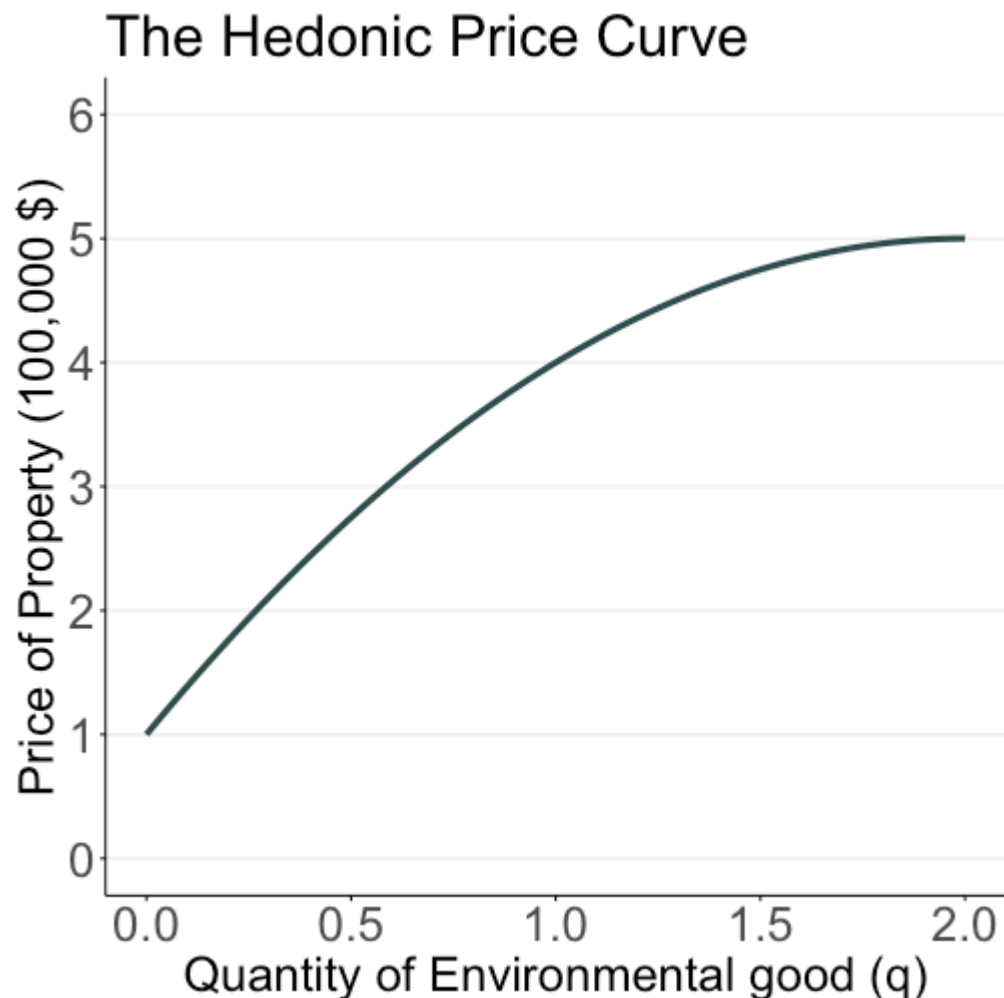
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P arises in equilibrium from the interaction of all buyers and sellers in the market

Here we will assume the supply of houses is fixed in the short run so the price curve arises solely from buyer behavior

The hedonic model: the price curve



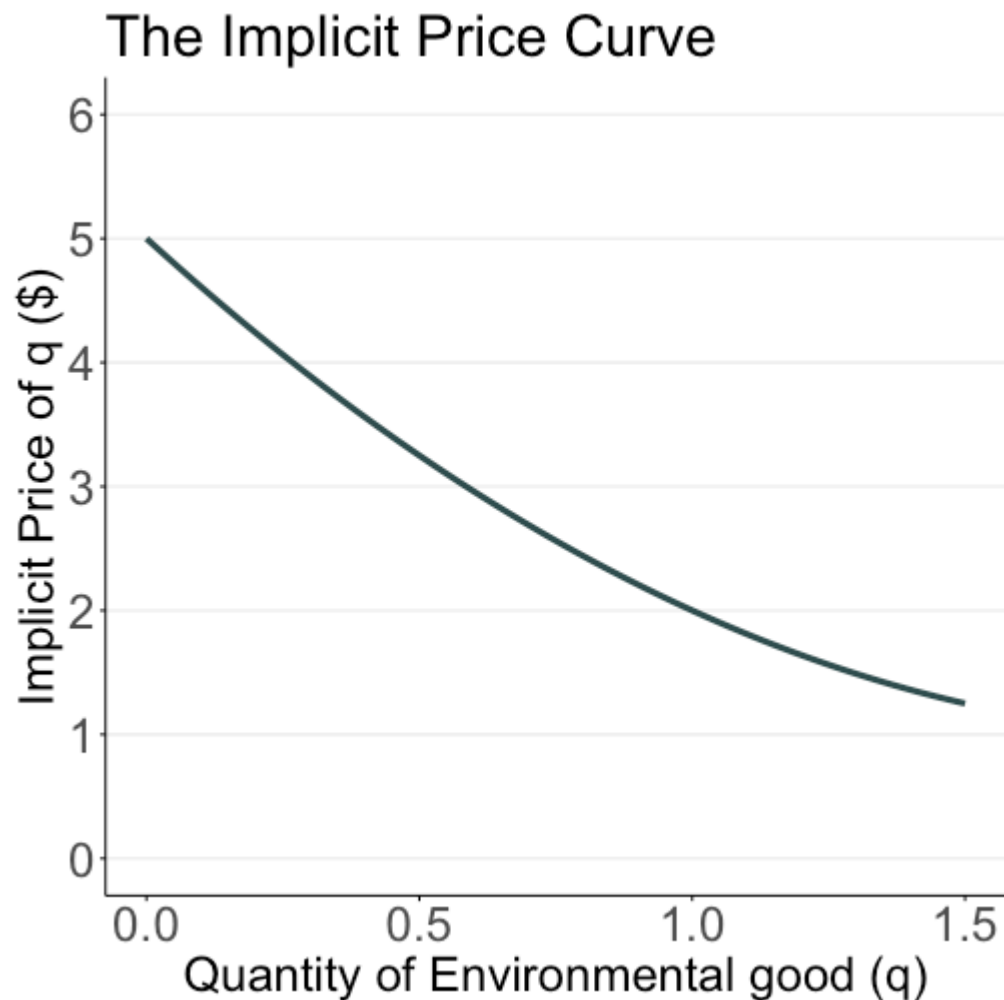
The hedonic price curve is $P(x, q)$

It's increasing in q (q is good) but at a decreasing rate (decreasing marginal utility)

This is holding x fixed

Analogous to regular demand curves holding income fixed

The hedonic model: the price curve



The implicit price curve for q is

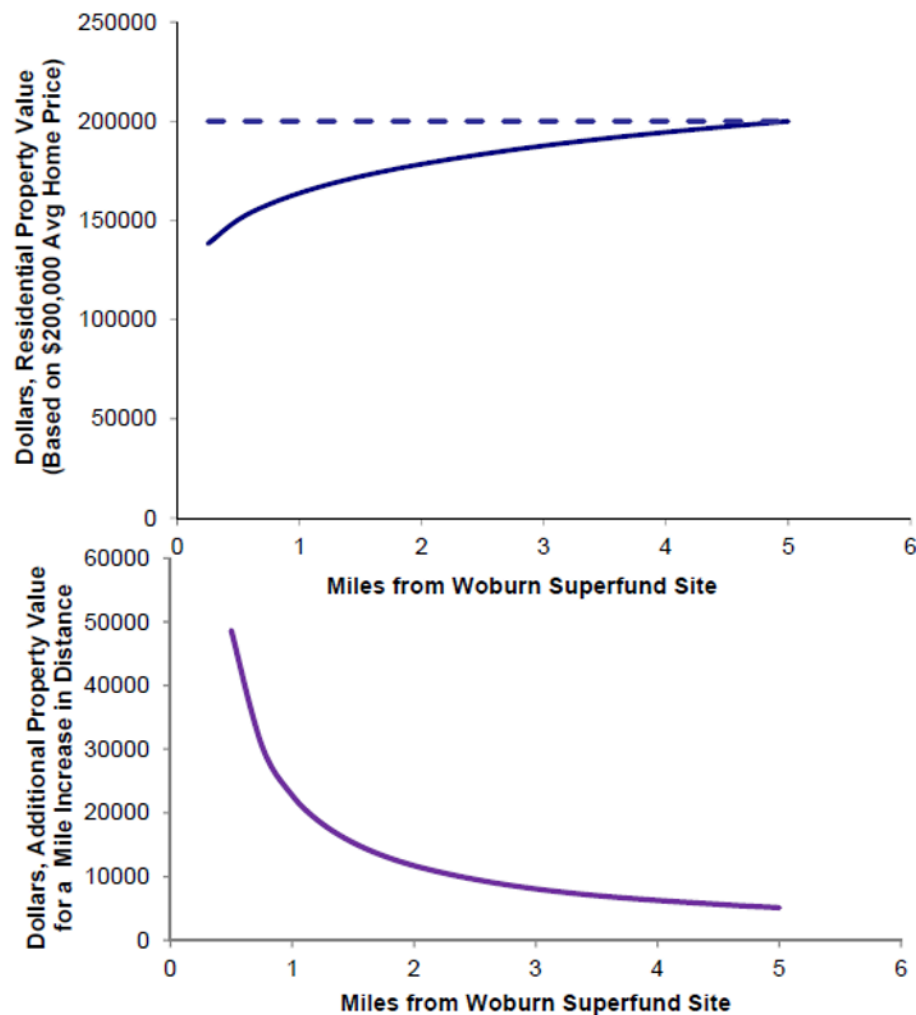
$$\frac{\partial P(x, q)}{\partial q}$$

It tells us how the price changes in q

It's positive, but downward sloping

This is effectively the environmental good demand curve

Price curve example



Total Value

“Marginal” Value
(one mile increment)

Source: Messer et al. *Env. and Res. Econ.* 2006

The hedonic model: consumer's choice problem

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- z is the numeraire good (spending on other private goods)
- y is income
- s is the set of the household's characteristics like family size, ages, etc

Choosing q

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We are thus also implicitly assuming q varies across space so that households can sort into areas they prefer

- q is really picking up **local** environmental goods

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For homeowners we are basically assuming they rent from themselves every year

The hedonic model: consumer's choice problem

$$\max_{x,q,z} U(x, q, z; s) \quad \text{subject to: } y = z + P(x, q)$$

Plug in the constraint for z to get:

$$\max_{x,q} U(x, q, \underbrace{y - P(x, q)}_z; s)$$

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$$\begin{aligned} \frac{\partial U}{\partial x_j} &= \frac{\partial U}{\partial z} \frac{\partial P}{\partial x_j} & j = 1, \dots, J & \quad (\text{house characteristics}) \\ \frac{\partial U}{\partial q} &= \frac{\partial U}{\partial z} \frac{\partial P}{\partial q} & & \quad (\text{environmental good}) \end{aligned}$$

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What does this mean?

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Recall from intro/intermediate micro: the MRS tells us how the household trades off q and z while keeping utility constant

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Recall from intro/intermediate micro: the MRS tells us how the household trades off q and z while keeping utility constant

Analogous to MRS = price ratio of two goods from intermediate micro, but here the price of z is just \$1

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How prices change in the environmental good, *holding all else constant*, tells us about WTP

The hedonic model in practice

What if all else isn't constant?

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What if all else isn't constant?

e.g. what if air quality improved in Syracuse because we are in a recession?

- Recessions make air quality better and prices higher (polluters aren't producing as much because demand is low)
- But recessions also decrease demand for houses and make prices lower (people are unemployed)

The hedonic model in practice

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If we see higher home prices as air quality improves (but we ignore the recession), then we are under-valuing air quality

The housing prices went up **despite** the recession!

The hedonic model in practice

How do we get around this?

Find a control city (e.g. Ithaca): houses that were also hit by the recession but **didn't** have an air quality improvement (why? maybe no polluters near by)

The hedonic model in practice

The change in prices of Ithaca houses tells us the impact of the recession, if we subtract it from the change in prices of Syracuse homes we get the effect of air quality alone!

- Syracuse price change - Ithaca price change = air quality effect
- (air quality effect + recession effect) - (recession effect) = air quality effect

We are **differencing out** other things that may affect housing prices

Housing prices and superfund clean up

Main question: How does superfund site clean up affects the housing price in the adjacent areas?

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How to answer it: Compare housing market outcomes in the areas surrounding the first 400 hazardous sites chosen for Superfund clean-ups to the areas surrounding the 290 sites that narrowly missed qualifying for these clean-ups

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Key idea: Any differences between housing values in these locations is most likely due to Superfund clean up, not other factors

Superfund location



Figure IIa
GEOGRAPHIC DISTRIBUTION OF HAZARDOUS WASTE SITES IN THE 1982 HRS SAMPLE
SITES WITH 1982 HRS SCORES EXCEEDING 28.5



Figure IIb
GEOGRAPHIC DISTRIBUTION OF HAZARDOUS WASTE SITES IN THE 1982 HRS SAMPLE
SITES WITH 1982 HRS SCORES BELOW 28.5

Regression

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They regress:

$$\log(2000 \text{ median home price})_c = \underbrace{\theta 1(\text{cleaned up in 2000})_c}_{= 1 \text{ if cleaned up, } = 0 \text{ if not}} + \beta \underbrace{X_c}_{\text{controls}} + \varepsilon_c$$

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— θ is telling us the **cost** of a superfund site to households

Superfund results: "quasi-experimental"

QUASI-EXPERIMENTAL ESTIMATES OF THE EFFECT OF NPL STATUS ON HOUSE PRICES, SAMPLES BASED ON THE 1982 HRS SAMPLE SITES

	<u>RD-Style Estimators</u>						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<u>A. Own Census Tract</u>							
1(NPL Status by 2000)	0.035	0.037	0.043	0.047	0.007	0.022	0.027
	(0.031)	(0.035)	(0.031)	(0.027)	(0.063)	(0.042)	(0.038)
<u>B. Adjacent Census Tracts</u>							
1(NPL Status by 2000)	0.071	0.066	0.012	0.015	-0.006	-0.002	0.001
	(0.031)	(0.035)	(0.029)	(0.022)	(0.056)	(0.035)	(0.035)
<u>C. 2-Mile Radius from Hazardous Waste Sites</u>							
1(NPL Status by 2000)	0.021	0.019	0.011	0.001	0.023	-0.018	-0.007
	(0.028)	(0.032)	(0.029)	(0.023)	(0.054)	(0.035)	(0.034)
Ho: > 0.138, P-Value	0.000	0.000	0.000	0.000	0.018	0.000	0.000

Top row of the last three columns are the important ones

Superfund results

Superfund cleanups had **economically and statistically insignificant effects** on property values, rental rates, housing supply, population, who lives near the site: 0.7-2.7% depending on the model

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Suggests the mean benefits of Superfund cleanup are small: at most, people demand only 3% lower housing prices to live near toxic waste!

Value of a statistical life (VSL)

How much should society spend, at the margin, to save a 'statistical life'?

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A statistical life is a probabilistic concept

VSL reflects willingness to pay for a reduction in the risk of death

VSL is more appropriately called the **value of mortality risk**

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How do you get a credible estimate of the VSL?

Value of a statistical life (VSL)

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How?

See tradeoffs people make between cost and safety

Value of a statistical life (VSL)

Some examples:

Value of a statistical life (VSL)

Some examples:

Driving speed

Value of a statistical life (VSL)

Some examples:

Driving speed

Vehicle choice

Value of a statistical life (VSL)

Some examples:

Driving speed

Vehicle choice

Wage-risk relationship

Value of a statistical life (VSL)

Some examples:

Driving speed

Vehicle choice

Wage-risk relationship

There's lots of studies, and lots of different answers

VSL

EPA recommends that the central estimate of \$7.4 million (\$2006), updated to the year of the analysis, be used in all benefits analyses that seek to quantify mortality risk reduction benefits regardless of the age, income, or other population characteristics of the affected population until revised guidance becomes available

VSL thought experiment

Suppose that individuals are willing to adopt a safety procedure, for which they have to give up 25 cents per hour, to reduce risk of on-the-job fatality by 1 in 10,000 (annual risk)

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This procedure would result in one fewer person dying on average

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$VSL = \$500 \times 10,000 = 5 \text{ million dollars}$

Estimating a hedonic wage function

We can estimate a **hedonic wage function**:

$$w_i = \alpha + \beta_1 H_i + \beta_2 X_i + \gamma_1 p_i + \gamma_2 q_i + \gamma_3 WC_i + \varepsilon_i$$

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$$w_i = \alpha + \beta_1 H_i + \beta_2 X_i + \gamma_1 p_i + \gamma_2 q_i + \gamma_3 WC_i + \varepsilon_i$$

w : wage

H : worker personal characteristics

X : job characteristics

p : **risk of death at the job**

q : non-fatal risk at the job

WC : workers' compensation benefits for injury

$\frac{\partial w}{\partial p}$ is the wage-risk trade off for marginal changes in risk

VSL from the hedonic wage function

Suppose:

- Wages were in thousands of dollars
- Risk is deaths per 10,000 people
- Coefficient on mortality risk p is $\gamma_1 = 0.4$

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- Wages were in thousands of dollars
- Risk is deaths per 10,000 people
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This implies an average WTP (reduced wage) of

VSL from the hedonic wage function

Suppose:

- Wages were in thousands of dollars
- Risk is deaths per 10,000 people
- Coefficient on mortality risk p is $\gamma_1 = 0.4$

This implies an average WTP (reduced wage) of 400 dollars to reduce risk by 1 in 10,000

VSL from the hedonic wage function

WTP (reduced wage) of 400 dollars to reduce risk by $1/10,000$

VSL from the hedonic wage function

WTP (reduced wage) of 400 dollars to reduce risk by $1/10,000$

This means that the VSL is:

VSL from the hedonic wage function

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This means that the VSL is:

$$VSL = \underbrace{(0.4 \times 1000)}_{\text{WTP to reduce risk by 1 in 10000}} \times 10,000 = 4 \text{ million dollars}$$

VSL from the hedonic wage function

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Suppose a policy reduces mortality risk by 1/10,000 for 60,000 people (saves 6 lives on average)

VSL from the hedonic wage function

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Suppose a policy reduces mortality risk by 1/10,000 for 60,000 people (saves 6 lives on average)

This policy has a value of:

$$400 * 60,000 = 24 \text{ million dollars}$$

VSL estimates

Exhibit 7-3 Value of Statistical Life Estimates (mean values in 1997 dollars)

Study	Method	Value of Statistical Life
Kneisner and Leeth (1991 - U.S.)	Labor Market	\$0.7 million
Smith and Gilbert (1984)	Labor Market	\$0.8 million
Dillingham (1985)	Labor Market	\$1.1 million
Buder (1983)	Labor Market	\$1.3 million
Miller and Guria (1991)	Contingent Valuation	\$1.5 million
Moore and Viscusi (1988)	Labor Market	\$3.0 million
Viscusi, Magat and Huber (1991)	Contingent Valuation	\$3.3 million
Marin and Psacharopoulos (1982)	Labor Market	\$3.4 million
Gegax et al. (1985)	Contingent Valuation	\$4.0 million
Kneisner and Leeth (1991 - Australia)	Labor Market	\$4.0 million
Gerking, de Haan and Schulze (1988)	Contingent Valuation	\$4.1 million
Cousineau, Lacroix and Girard (1988)	Labor Market	\$4.4 million
Jones-Lee (1989)	Contingent Valuation	\$4.6 million
Dillingham (1985)	Labor Market	\$4.7 million
Viscusi (1978, 1979)	Labor Market	\$5.0 million
R.S. Smith (1976)	Labor Market	\$5.6 million
V.K. Smith (1976)	Labor Market	\$5.7 million
Olson (1981)	Labor Market	\$6.3 million
Viscusi (1981)	Labor Market	\$7.9 million
R.S. Smith (1974)	Labor Market	\$8.7 million
Moore and Viscusi (1988)	Labor Market	\$8.8 million
Kneisner and Leeth (1991 - Japan)	Labor Market	\$9.2 million
Herzog and Schlottman (1987)	Labor Market	\$11.0 million
Leigh and Folsom (1984)	Labor Market	\$11.7 million
Leigh (1987)	Labor Market	\$12.6 million
Garen (1988)	Labor Market	\$16.3 million
Derived from EPA (1997) and Viscusi (1992).		